

1. Language Modelling / Next Token Prediction

A language model estimates the probability of a sequence of words. In next-token prediction, the model learns to predict the most likely next token given all previous tokens, capturing dependencies across time. While in training, we minimize the cross-entropy between predictions and true tokens, which ensures the model assigns a higher probability to real text.

2. Self-Attention Mechanism

Self-Attention allows each token to look at other tokens in the same sequence to gather contextual information. For each token, key, and value, vectors are computed. The attention weights determine how much information to pull from other positions. This helps the model capture long-range dependencies without having context windows.

3. Model Architecture and Implementation

Used `nn.Embedding` layer that maps to each input token ID to a dense vector of $d_{model} = 256$

Added sinusoidal positional encodings

Combined them in `EmbeddingWithPositions`

Encoder used 4 transformer layers, 6 attention heads each

Used LayerNorm, GELU activation, dropout = 0.1

Output head: single layer tied to token embeddings

Optimizer: Adam(lr = 2e-4, betas = (0.9, 0.95), weight_decay = 0.1)

Gradient clipping (max_norm = 0.5)

Warmup and cosine LS scheduler that was set to 10%

CrossEntropyLoss(ignore_index = 100)

4. Model Summary

Component	Description
Token embedding	384-dimensional vectors
Positional encoding	Sinusoidal (fixed)
Heads	6 self-attention heads
Layers	4 transformer encoder layers
Dropout	0.1
Output head	Linear layer to vocab size
Parameters	~22M
Optimizer	AdamW(lr = 2e-4)
Loss	CrossEntropy(ignore_index = -100)
Task	Next-token prediction
Dataset	Penn Treebank

5. Results and Perplexity

Preplexity decreases steadily across epochs, showing there's effective learning. The final test, the perplexity (=101), aligns with expectations for an embedding-only model.

Table 1: Training Progress (Test 3)

Epoch	Train Loss	Train PPL	Val Loss	Val PPL
1	15.85	7,619,375.53	9.54	13,876.38
2	9.27	10,621.20	7.27	1,438.41
3	7.05	1,149.78	6.14	462.94
4	5.91	369.97	5.56	258.71
5	5.33	207.27	5.24	189.32
6	5.02	151.74	5.07	159.05
7	4.84	126.43	4.96	142.38
8	4.72	112.52	4.89	132.76
9	4.64	103.76	4.83	124.96
10	4.58	97.84	4.80	121.30